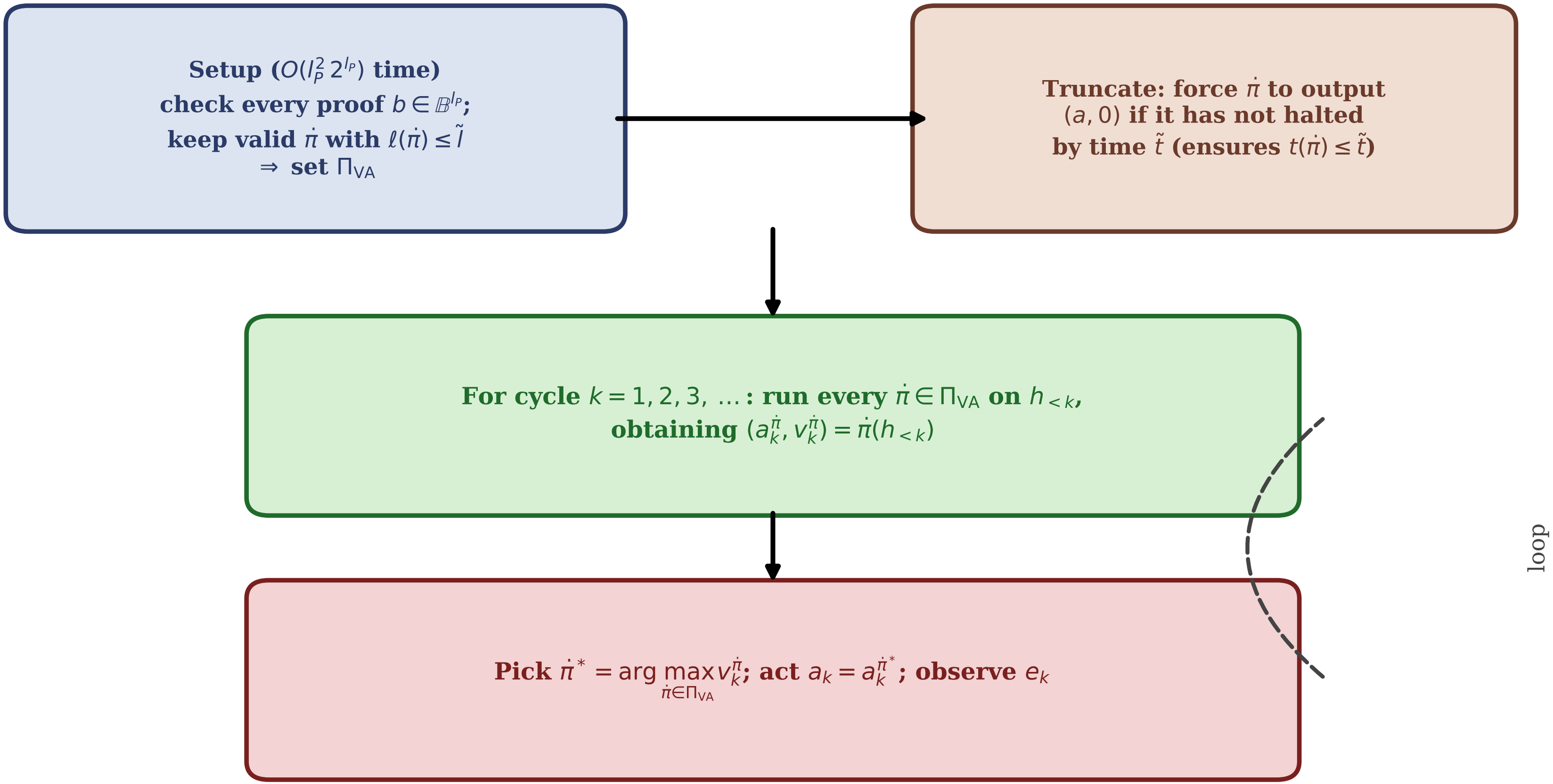

Per-cycle cost $O(t_{\mathrm{cycle}}) = O(2^{\tilde{l}} \cdot \tilde{t})$ — independent of the number of elapsed cycles k